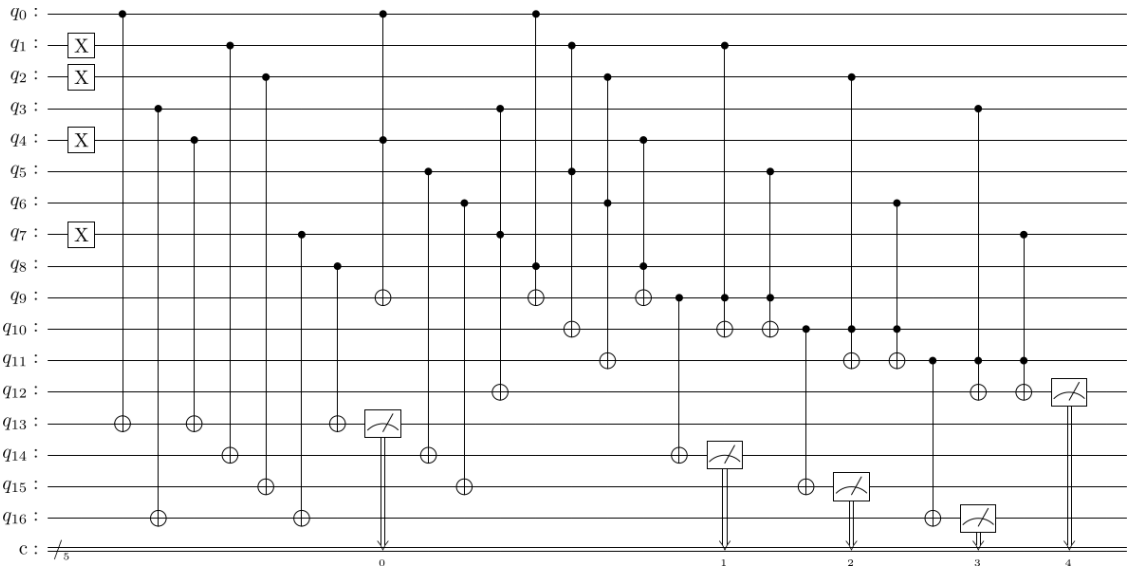
Lab 7: Multi-Bit Ripple-Carry Adders Workshop

1 4-bit ripple-carry full-adder

1.1 Circuit

The flowing circuit is implemented to compute the sum of 2 4-bit numbers (the given one is for 0110 + 1001),



q_0 through q_3 are used to store A. q_4 through q_7 are used to store B. q_8 through q_{12} are used to store the carry from each qubit addition, where q_8 is C_{in}. q_{13} through q_{16} are used to store the sum from each qubit addition. For each carry-bit, a CCNOT gate is used on the A-bit and B-bit, the A-bit and the prevous carry-bit, and the B-bit and the prevous carry-bit. For each sum-bit a CNOT gate is used on the A-bit, the B-bit and the carry-bit.

1.2 Truth Table

The truth table for each possible bit combination is given. The compare row compares adding A and B to the integer representation of the output bit string including the carry.

A	**B**	**C_{in}**	**Sum**	**C_{out}**	**Compare**
0000	0000	0	0000	0	0 = 0
0000	0001	0	0001	0	1 = 1
0000	0010	0	0010	0	2 = 2
0000	0011	0	0011	0	3 = 3
0000	0100	0	0100	0	4 = 4
0000	0101	0	0101	0	5 = 5
0000	0110	0	0110	0	6 = 6
0000	0111	0	0111	0	7 = 7
0000	1000	0	1000	0	8 = 8
0000	1001	0	1001	0	9 = 9
0000	1010	0	1010	0	10 = 10
0000	1011	0	1011	0	11 = 11
0000	1100	0	1100	0	12 = 12

0000	1101	0	1101	0	13 = 13
0000	1110	0	1110	0	14 = 14
0000	1111	0	1111	0	15 = 15
0001	0000	0	0001	0	1 = 1
0001	0001	0	0010	0	2 = 2
0001	0010	0	0011	0	3 = 3
0001	0011	0	0100	0	4 = 4
0001	0100	0	0101	0	5 = 5
0001	0101	0	0110	0	6 = 6
0001	0110	0	0111	0	7 = 7
0001	0111	0	1000	0	8 = 8
0001	1000	0	1001	0	9 = 9
0001	1001	0	1010	0	10 = 10
0001	1010	0	1011	0	11 = 11
0001	1011	0	1100	0	12 = 12
0001	1100	0	1101	0	13 = 13
0001	1101	0	1110	0	14 = 14
0001	1110	0	1111	0	15 = 15
0001	1111	0	0000	1	16 = 16
0010	0000	0	0010	0	2 = 2
0010	0001	0	0011	0	3 = 3
0010	0010	0	0100	0	4 = 4
0010	0011	0	0101	0	5 = 5
0010	0100	0	0110	0	6 = 6
0010	0101	0	0111	0	7 = 7
0010	0110	0	1000	0	8 = 8
0010	0111	0	1001	0	9 = 9
0010	1000	0	1010	0	10 = 10
0010	1001	0	1011	0	11 = 11
0010	1010	0	1100	0	12 = 12
0010	1011	0	1101	0	13 = 13
0010	1100	0	1110	0	14 = 14
0010	1101	0	1111	0	15 = 15
0010	1110	0	0000	1	16 = 16
0010	1111	0	0001	1	17 = 17
0011	0000	0	0011	0	3 = 3
0011	0001	0	0100	0	4 = 4
0011	0010	0	0101	0	5 = 5
0011	0011	0	0110	0	6 = 6
0011	0100	0	0111	0	7 = 7
0011	0101	0	1000	0	8 = 8
0011	0110	0	1001	0	9 = 9
0011	0111	0	1010	0	10 = 10
0011	1000	0	1011	0	11 = 11
0011	1001	0	1100	0	12 = 12
0011	1010	0	1101	0	13 = 13
0011	1011	0	1110	0	14 = 14
0011	1100	0	1111	0	15 = 15
0011	1101	0	0000	1	16 = 16
0011	1110	0	0001	1	17 = 17

0011	1111	0	0010	1	18 = 18
0100	0000	0	0100	0	4 = 4
0100	0001	0	0101	0	5 = 5
0100	0010	0	0110	0	6 = 6
0100	0011	0	0111	0	7 = 7
0100	0100	0	1000	0	8 = 8
0100	0101	0	1001	0	9 = 9
0100	0110	0	1010	0	10 = 10
0100	0111	0	1011	0	11 = 11
0100	1000	0	1100	0	12 = 12
0100	1001	0	1101	0	13 = 13
0100	1010	0	1110	0	14 = 14
0100	1011	0	1111	0	15 = 15
0100	1100	0	0000	1	16 = 16
0100	1101	0	0001	1	17 = 17
0100	1110	0	0010	1	18 = 18
0100	1111	0	0011	1	19 = 19
0101	0000	0	0101	0	5 = 5
0101	0001	0	0110	0	6 = 6
0101	0010	0	0111	0	7 = 7
0101	0011	0	1000	0	8 = 8
0101	0100	0	1001	0	9 = 9
0101	0101	0	1010	0	10 = 10
0101	0110	0	1011	0	11 = 11
0101	0111	0	1100	0	12 = 12
0101	1000	0	1101	0	13 = 13
0101	1001	0	1110	0	14 = 14
0101	1010	0	1111	0	15 = 15
0101	1011	0	0000	1	16 = 16
0101	1100	0	0001	1	17 = 17
0101	1101	0	0010	1	18 = 18
0101	1110	0	0011	1	19 = 19
0101	1111	0	0100	1	20 = 20
0110	0000	0	0110	0	6 = 6
0110	0001	0	0111	0	7 = 7
0110	0010	0	1000	0	8 = 8
0110	0011	0	1001	0	9 = 9
0110	0100	0	1010	0	10 = 10
0110	0101	0	1011	0	11 = 11
0110	0110	0	1100	0	12 = 12
0110	0111	0	1101	0	13 = 13
0110	1000	0	1110	0	14 = 14
0110	1001	0	1111	0	15 = 15
0110	1010	0	0000	1	16 = 16
0110	1011	0	0001	1	17 = 17
0110	1100	0	0010	1	18 = 18
0110	1101	0	0011	1	19 = 19
0110	1110	0	0100	1	20 = 20
0110	1111	0	0101	1	21 = 21
0111	0000	0	0111	0	7 = 7

0111	0001	0	1000	0	8 = 8
0111	0010	0	1001	0	9 = 9
0111	0011	0	1010	0	10 = 10
0111	0100	0	1011	0	11 = 11
0111	0101	0	1100	0	12 = 12
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0111	0111	0	1110	0	14 = 14
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0111	1001	0	0000	1	16 = 16
0111	1010	0	0001	1	17 = 17
0111	1011	0	0010	1	18 = 18
0111	1100	0	0011	1	19 = 19
0111	1101	0	0100	1	20 = 20
0111	1110	0	0101	1	21 = 21
0111	1111	0	0110	1	22 = 22
1000	0000	0	1000	0	8 = 8
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1000	0010	0	1010	0	10 = 10
1000	0011	0	1011	0	11 = 11
1000	0100	0	1100	0	12 = 12
1000	0101	0	1101	0	13 = 13
1000	0110	0	1110	0	14 = 14
1000	0111	0	1111	0	15 = 15
1000	1000	0	0000	1	16 = 16
1000	1001	0	0001	1	17 = 17
1000	1010	0	0010	1	18 = 18
1000	1011	0	0011	1	19 = 19
1000	1100	0	0100	1	20 = 20
1000	1101	0	0101	1	21 = 21
1000	1110	0	0110	1	22 = 22
1000	1111	0	0111	1	23 = 23
1001	0000	0	1001	0	9 = 9
1001	0001	0	1010	0	10 = 10
1001	0010	0	1011	0	11 = 11
1001	0011	0	1100	0	12 = 12
1001	0100	0	1101	0	13 = 13
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1001	1100	0	0101	1	21 = 21
1001	1101	0	0110	1	22 = 22
1001	1110	0	0111	1	23 = 23
1001	1111	0	1000	1	24 = 24
1010	0000	0	1010	0	10 = 10
1010	0001	0	1011	0	11 = 11
1010	0010	0	1100	0	12 = 12

1010	0011	0	1101	0	13 = 13
1010	0100	0	1110	0	14 = 14
1010	0101	0	1111	0	15 = 15
1010	0110	0	0000	1	16 = 16
1010	0111	0	0001	1	17 = 17
1010	1000	0	0010	1	18 = 18
1010	1001	0	0011	1	19 = 19
1010	1010	0	0100	1	20 = 20
1010	1011	0	0101	1	21 = 21
1010	1100	0	0110	1	22 = 22
1010	1101	0	0111	1	23 = 23
1010	1110	0	1000	1	24 = 24
1010	1111	0	1001	1	25 = 25
1011	0000	0	1011	0	11 = 11
1011	0001	0	1100	0	12 = 12
1011	0010	0	1101	0	13 = 13
1011	0011	0	1110	0	14 = 14
1011	0100	0	1111	0	15 = 15
1011	0101	0	0000	1	16 = 16
1011	0110	0	0001	1	17 = 17
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1101	0011	0	0000	1	16 = 16
1101	0100	0	0001	1	17 = 17

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1101	0111	0	0100	1	20 = 20
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1101	1001	0	0110	1	22 = 22
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0000	1110	1	1111	0	$14 = 15$
0000	1111	1	0000	1	$15 = 16$
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0011	1000	1	1100	0	$11 = 12$

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0011	1011	1	1111	0	14 = 15
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0011	1101	1	0001	1	16 = 17
0011	1110	1	0010	1	17 = 18
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0100	0010	1	0111	0	6 = 7
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0100	1001	1	1110	0	13 = 14
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1000	1110	1	0111	1	22 = 23
1000	1111	1	1000	1	23 = 24
1001	0000	1	1010	0	9 = 10
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1001	1111	1	1001	1	24 = 25
1010	0000	1	1011	0	10 = 11
1010	0001	1	1100	0	11 = 12
1010	0010	1	1101	0	12 = 13
1010	0011	1	1110	0	13 = 14
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1100	1001	1	0110	1	21 = 22
1100	1010	1	0111	1	22 = 23
1100	1011	1	1000	1	23 = 24
1100	1100	1	1001	1	24 = 25
1100	1101	1	1010	1	25 = 26
1100	1110	1	1011	1	26 = 27

1100	1111	1	1100	1	27 = 28
1101	0000	1	1110	0	13 = 14
1101	0001	1	1111	0	14 = 15
1101	0010	1	0000	1	15 = 16
1101	0011	1	0001	1	16 = 17
1101	0100	1	0010	1	17 = 18
1101	0101	1	0011	1	18 = 19
1101	0110	1	0100	1	19 = 20
1101	0111	1	0101	1	20 = 21
1101	1000	1	0110	1	21 = 22
1101	1001	1	0111	1	22 = 23
1101	1010	1	1000	1	23 = 24
1101	1011	1	1001	1	24 = 25
1101	1100	1	1010	1	25 = 26
1101	1101	1	1011	1	26 = 27
1101	1110	1	1100	1	27 = 28
1101	1111	1	1101	1	28 = 29
1110	0000	1	1111	0	14 = 15
1110	0001	1	0000	1	15 = 16
1110	0010	1	0001	1	16 = 17
1110	0011	1	0010	1	17 = 18
1110	0100	1	0011	1	18 = 19
1110	0101	1	0100	1	19 = 20
1110	0110	1	0101	1	20 = 21
1110	0111	1	0110	1	21 = 22
1110	1000	1	0111	1	22 = 23
1110	1001	1	1000	1	23 = 24
1110	1010	1	1001	1	24 = 25
1110	1011	1	1010	1	25 = 26
1110	1100	1	1011	1	26 = 27
1110	1101	1	1100	1	27 = 28
1110	1110	1	1101	1	28 = 29
1110	1111	1	1110	1	29 = 30
1111	0000	1	0000	1	15 = 16
1111	0001	1	0001	1	16 = 17
1111	0010	1	0010	1	17 = 18
1111	0011	1	0011	1	18 = 19
1111	0100	1	0100	1	19 = 20
1111	0101	1	0101	1	20 = 21
1111	0110	1	0110	1	21 = 22
1111	0111	1	0111	1	22 = 23
1111	1000	1	1000	1	23 = 24
1111	1001	1	1001	1	24 = 25
1111	1010	1	1010	1	25 = 26
1111	1011	1	1011	1	26 = 27
1111	1100	1	1100	1	27 = 28
1111	1101	1	1101	1	28 = 29
1111	1110	1	1110	1	29 = 30
1111	1111	1	1111	1	30 = 31

1.3 Gate Count, Depth, and Scalability

Running qc.count_ops() on the quantum circuit results in: ('cx', 12), ('ccx', 12), ('measure', 5), ('x', 4). Running qc.depth() results in 15. Compared to the classical circuit (not counting the measure or x gates needed to prepare the states), a total of 24 gates is slightly more than 20 needed for a full adder, meaning just one extra gate per bit is need for the quatum reversability. On top of the extra gate each bit will require more qubits, as we need to store the carry to ensure reproducability. When scaling to more bits, each carry bit will need 3 Toffoli gates [CCNOT(A,B,C_{out}),CCNOT(A,C_{in},C_{out}),CCNOT(B,C_{in},C_{out})], so the gate grows as $3n$, where n is the number of bits.

1.4 Code

The code below was used to generate the truth table, the circuit, and to implement the full adder. It should work for any integer number of bits ($n \geq 1$).

```
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Statevector, state_fidelity
from qiskit_aer import AerSimulator
from qiskit.visualization import plot_bloch_multivector, plot_state_qsphere, plot_histogram
import matplotlib.pyplot as plt
import numpy as np
sim = AerSimulator()

def run_full_adder_n_bit(A,B,Cin, numOfBits):
    qc = QuantumCircuit(4*numOfBits+1,numOfBits+1)
    A.reverse()
    B.reverse()

    # Initialize qubits
    A_bits = []
    B_bits = []
    C_bits = []
    Sum_bits = []
    for i in range(0,numOfBits):
        A_bits.append(i)
        B_bits.append(i + numOfBits)
        C_bits.append(i + (2*numOfBits))
        Sum_bits.append(i + 1 +(3*numOfBits))
    C_bits.append(3*numOfBits)

    # Initialize qubit inputs
    for i in range(0,numOfBits):
        if A[i]: qc.x(A_bits[i])
        if B[i]: qc.x(B_bits[i])

    if Cin: qc.x(C_bits[0])


    for i in range(0,numOfBits):
        # Add A,B and Cin
        qc.cx(A_bits[i],Sum_bits[i])
        qc.cx(B_bits[i],Sum_bits[i])
        qc.cx(C_bits[i],Sum_bits[i])

```

```
        # Compute the carry bit
        qc.ccx(A_bits[i],B_bits[i],C_bits[i+1])
        qc.ccx(A_bits[i],C_bits[i],C_bits[i+1])
        qc.ccx(B_bits[i],C_bits[i],C_bits[i+1])

    # Measure the sums and last carry
    for i in range(0,numOfBits):
        qc.measure(Sum_bits[i],i)
    qc.measure(C_bits[numOfBits],numOfBits)

    compiled = transpile(qc, sim)
    result = sim.run(compiled, shots = 1).result()

    bits = list(result.get_counts().keys())[0]
    A.reverse()
    B.reverse()
    cout = bits[0]
    sum = bits[1:]
    return sum,cout, qc

def full_adder_sum_checker(n):
    print(f"{"A"}", "+", f"{"B"}", "+", f"{"C"}", "=", f"Out")
    for cin in [0,1]:
        for a in range(0, 2**n):
            a_bits = [int(a_bit) for a_bit in format(a, f"0{n}b")]
            for b in range(0, 2**n):
                b_bits = [int(b_bit) for b_bit in format(b, f"0{n}b")]
                sum, cout, _ = run_full_adder_n_bit(a_bits,b_bits,int(cin), int(n))

                a_string = format(a, f"0{n}b")
                b_string = format(b, f"0{n}b")
                print(f"{a}", "+ " f"{b}", "+", f"{cin}", "= " f"{int(sum + cout,2)}")

def full_adder_truth_table_generator(n):
    # Generate every combination of A, B, and Cin
    # print(f"{"A":^{n}}", f"{"B":^{n}}", f"{"C"}", f"{"S":^{n}}",f"{"C"}")
    for cin in [0,1]:
        for a in range(0, 2**n):
            a_bits = [int(a_bit) for a_bit in format(a, f"0{n}b")]
            for b in range(0, 2**n):
                b_bits = [int(b_bit) for b_bit in format(b, f"0{n}b")]
                sum, cout,_ = run_full_adder_n_bit(a_bits,b_bits,int(cin), int(n))
                a_string = format(a, f"0{n}b")
                b_string = format(b, f"0{n}b")
                # print(f"{a_string}", f"{b_string}", f"{cin}", f"{sum}", f"{cout}")
                print(f"{a_string}", f"{"&"}", f"{b_string}", f"{"&"}", f"{cin}", f"{"&"}", f"{sum}", f
```